

Brain Tumor Detection Using Convolutional Neural Network

Software requirement Specification Document

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S23BINFT1M01142

SESSION : 2023 – 2027 Information Technology /The Islamia University Of Bahawalpur

Revision History

Date	Description	Author	Comments
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Document Approval

Signature	Printed Name	Tittle	Date
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SRS STATEMENT

This Software Requirements Specification (SRS) document provides a comprehensive description of the system titled “**Brain Tumor Detection using Convolutional Neural Network (CNN)**”. It outlines the functional and non-functional requirements, system architecture, design constraints, and operational environment of the proposed system.

The purpose of this document is to serve as a reference for developers, project supervisors, and stakeholders to understand the system’s behavior and requirements before implementation. It ensures that all system features are clearly defined and helps in minimizing development errors and misunderstandings.

This SRS acts as a formal agreement between stakeholders and the development team and will be used as a guideline throughout the system development life cycle.

DECLARATION

I hereby declare that this project report titled “**Brain Tumor Detection using Convolutional Neural Network (CNN)**” is my original work and has been carried out by me under the supervision of my project supervisor.

This work has not been submitted previously to any other university or institution for the award of any degree or diploma. All sources of information used in this project have been properly acknowledged and referenced.

ABSTRACT

Brain tumors are among the most critical and life-threatening diseases that require early and accurate diagnosis for effective treatment. This project presents an automated system for detecting brain tumors from MRI images using a deep learning approach known as Convolutional Neural Networks (CNN).

The proposed system processes MRI images through several stages, including image preprocessing, feature extraction, and classification. The CNN model is trained on a dataset of brain MRI scans to distinguish between tumor and non-tumor images with high accuracy.

This system aims to assist medical professionals by providing fast, reliable, and accurate predictions, thereby reducing manual effort and the chances of human error. The results demonstrate that the CNN-based approach is effective in medical image analysis and can be further extended for advanced tumor classification and real-time applications.

Overall, this project highlights the potential of artificial intelligence in improving healthcare diagnostics and decision-making processes.

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4. Introduction

4.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive and detailed description of the Brain Tumor Detection System using Convolutional Neural Networks (CNN). The purpose of this system is to assist medical professionals in diagnosing brain tumors from MRI (Magnetic Resonance Imaging) scans using artificial intelligence techniques. The system aims to improve diagnostic accuracy, reduce human error, and provide faster results.

Early detection of brain tumors is critical for effective treatment planning and improving patient survival rates. Traditional diagnosis methods rely heavily on radiologists, which may be time-consuming and subject to variability. This system introduces automation using deep learning to enhance decision-making in medical imaging.

4.2 Scope

The system will:

- Accept MRI brain scan images as input
- Perform preprocessing such as resizing, normalization, and noise removal
- Use a trained CNN model to detect tumors
- Classify images into Tumor or No Tumor categories
- Provide prediction confidence scores
- Display results via graphical interface
- Store prediction history for future analysis

The system is intended for hospitals, clinics, diagnostic laboratories, and research institutions.

4.3 Definitions, Acronyms, Abbreviations

- CNN: Convolutional Neural Network
- MRI: Magnetic Resonance Imaging
- AI: Artificial Intelligence
- ML: Machine Learning
- GUI: Graphical User Interface
- SRS: Software Requirements Specification

4.4 Document Overview

This document includes system architecture, functional and non-functional requirements, diagrams, and implementation details necessary for development and deployment.

5. Overall Description

5.1 Product Perspective

This is a standalone AI-based application that may later integrate with hospital management systems. It follows a modular architecture with clear separation between UI, processing, and model layers.

5.2 Product Functions

- Upload MRI images
- Preprocess input data
- Extract features using CNN
- Classify tumor presence
- Display and store results

5.3 User Classes and Characteristics

- Doctors: Require high accuracy and easy interpretation
- Technicians: Operate system and upload images
- Researchers: Analyze performance and improve model

5.4 Operating Environment

- OS: Windows, Linux
- Language: Python
- Libraries: TensorFlow, Keras, OpenCV, NumPy
- Hardware: CPU (minimum), GPU (recommended)

5.5 Constraints

- Requires labeled MRI dataset
- High computational cost for training
- Accuracy depends on dataset quality

5.6 Assumptions and Dependencies

- Users have basic computer knowledge
- Reliable dataset is available

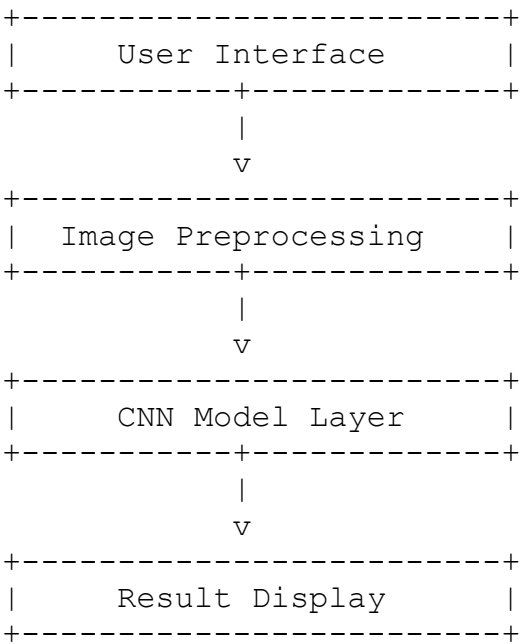
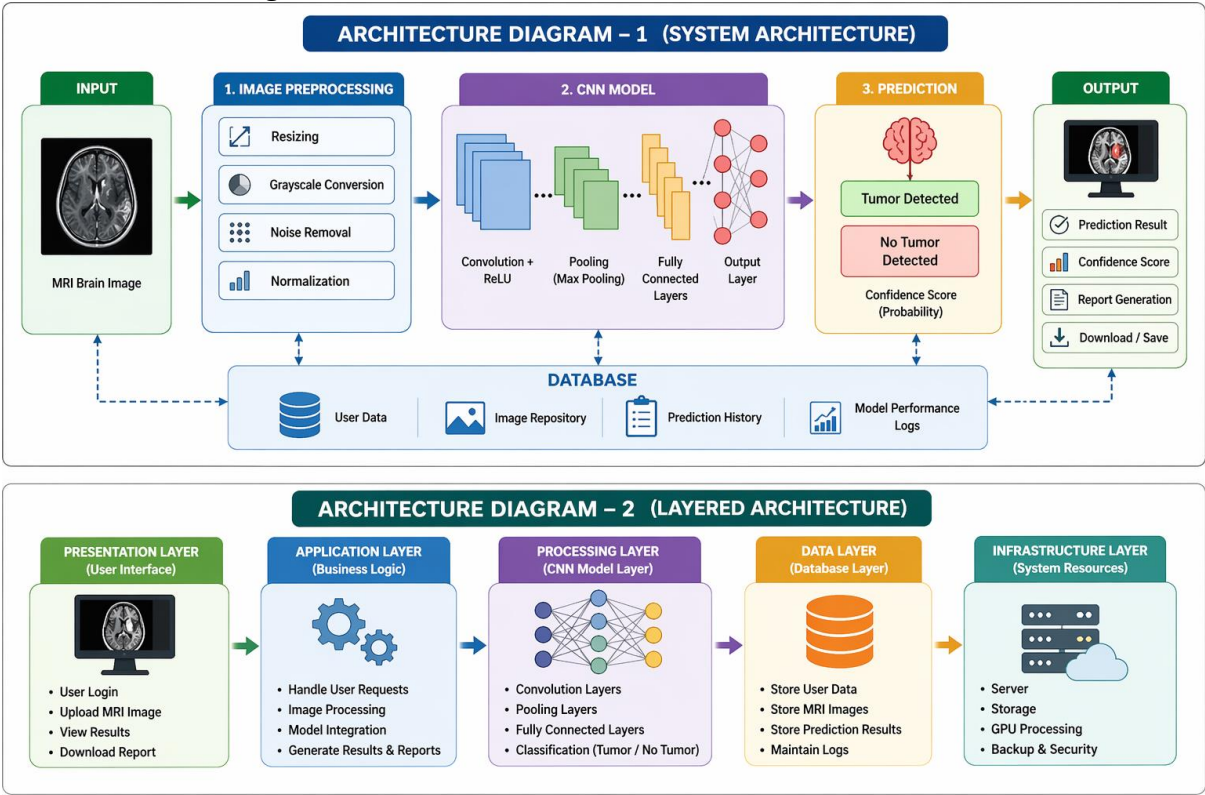
6. System Architecture

6.1 Architecture Overview

The system follows layered architecture:

1. Presentation Layer
2. Application Layer
3. Model Layer
4. Data Layer

6.2 Architecture Diagram

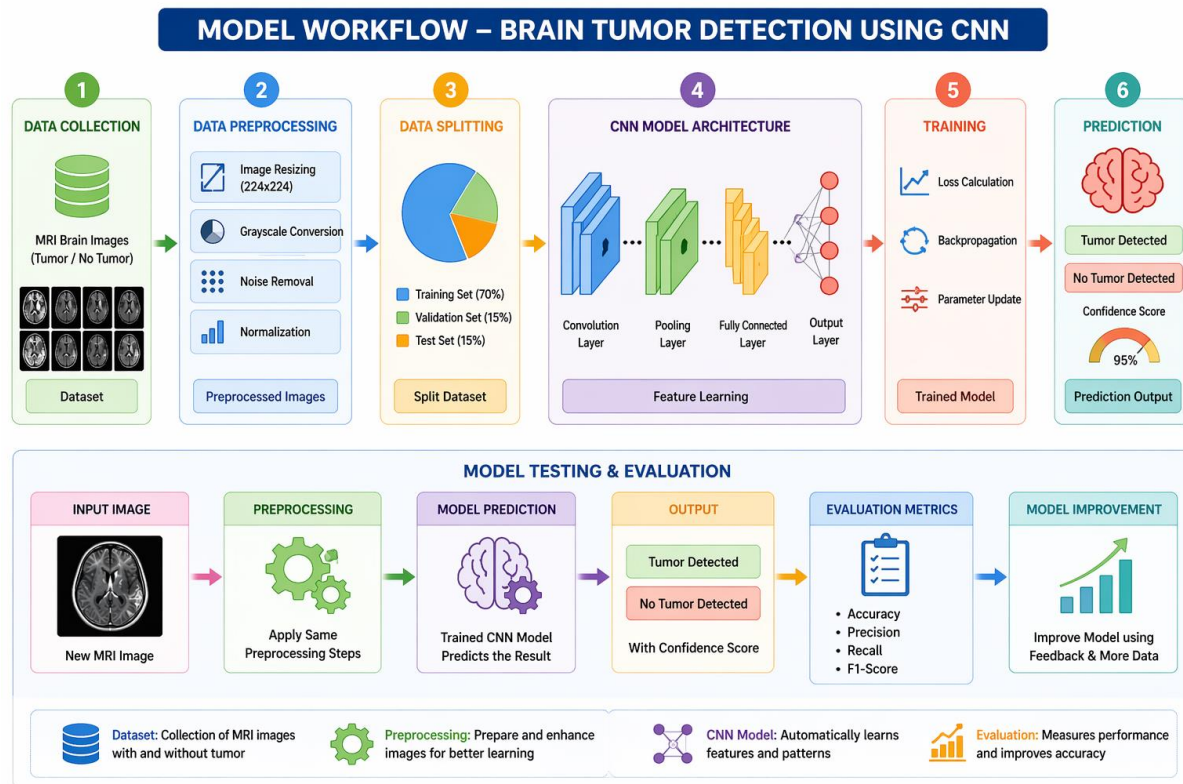


7. CNN Model Description

The CNN model consists of multiple layers:

1. Convolution Layer: Extracts features
2. ReLU Activation: Adds non-linearity
3. Pooling Layer: Reduces dimensions
4. Fully Connected Layer: Classification

7.1 Model Workflow



```
Input Image
↓
Convolution + ReLU
↓
Pooling
↓
Flatten
↓
Fully Connected Layer
↓
Output (Tumor / No Tumor)
```

7.2 Training Details

- Loss Function: Binary Crossentropy
- Optimizer: Adam
- Epochs: 20–50
- Accuracy Target: >90%

8. System Features

8.1 Image Upload Module

- Upload MRI images
- Validate format and size

8.2 Preprocessing Module

- Resize images
- Normalize pixel values
- Remove noise

8.3 Prediction Module

- Run CNN model
- Generate prediction

8.4 Output Module

- Display result
- Show confidence score

8.5 Report Generation Module

Description:

Generates reports based on prediction results.

Features:

- Downloadable report (PDF format)
- Includes image, result, and confidence score
- Timestamp of analysis

1 Functional Requirements:

- The system shall generate a report
- The system shall allow report download

Data Storage Module

Description:

Stores user data and prediction results.

Features:

- Save prediction history
- Store uploaded images
- Maintain logs

Functional Requirements:

- The system shall store prediction results
- The system shall allow retrieval of past records

Error Handling Module

Description:

Ensures smooth system operation by handling errors.

Features:

- Input validation errors
- System crash prevention
- User-friendly error messages

Functional Requirements:

- The system shall display meaningful error messages
- The system shall prevent system failure

Performance Monitoring Module (Advanced)

Description:

Tracks system performance and model accuracy.

Features:

- Accuracy tracking
- Processing time monitoring
- Model evaluation metrics

Functional Requirements:

- The system shall monitor prediction performance
- The system shall display accuracy metric

9. Functional Requirements

1. System shall allow image upload
2. System shall validate image
3. System shall preprocess data
4. System shall run CNN model
5. System shall classify tumor presence
6. System shall display results
7. System shall store predictions
8. System shall allow report download
9. System shall handle errors
10. System shall maintain logs

10. Non-Functional Requirements

10.1 Performance

- Prediction time < 5 seconds
- High accuracy (>90%)

10.2 Security

- Protect patient data
- Secure storage

10.3 Usability

- User-friendly interface
- Minimal training required

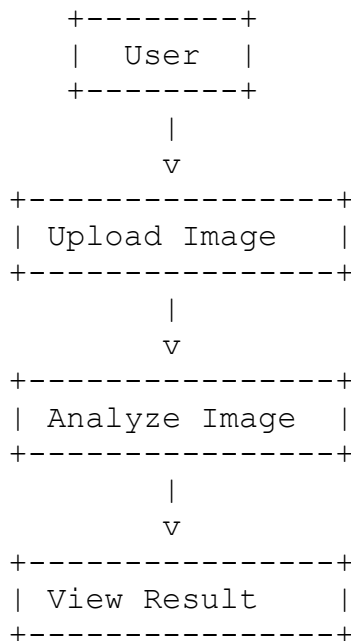
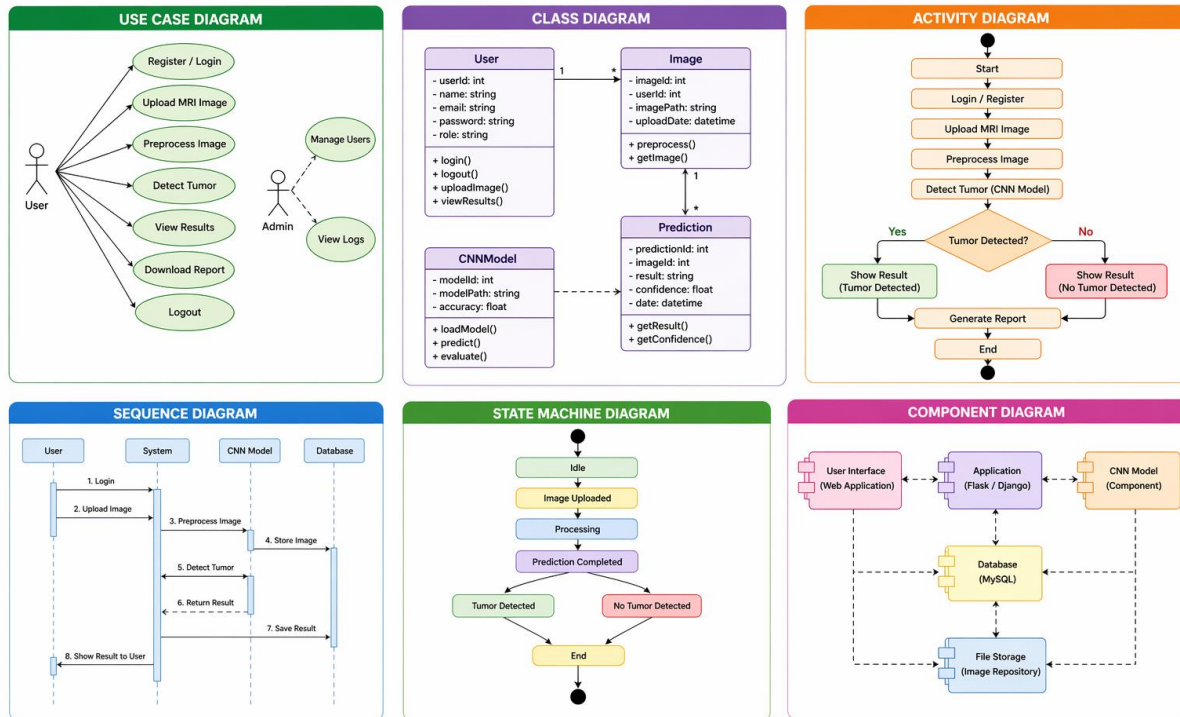
10.4 Reliability

- System uptime > 95%

11. UML Diagrams

11. Use Case Diagram

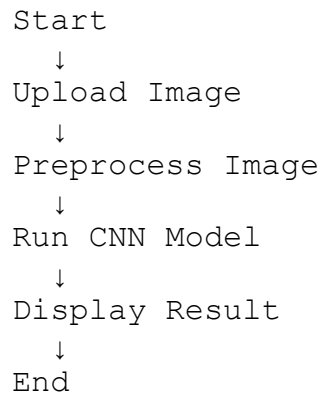
UML DIAGRAMS – BRAIN TUMOR DETECTION SYSTEM



11.2 Sequence Diagram

User → UI → Preprocessing → CNN Model → Output

11.3 Activity Diagram



12. Data Flow Diagram (DFD)



13. Database Design (Optional)

Table: Predictions

- ID
- Image Name
- Result
- Confidence Score
- Date

14. Testing Strategy

14.1 Unit Testing

Test each module individually

14.2 Integration Testing

Ensure modules work together

14.3 System Testing

Test full system functionality

14.4 Performance Testing

Check speed and accuracy

15. Risk Analysis

- Poor dataset quality
- Model overfitting
- Hardware limitations

16. Future Enhancements

- Tumor type classification
- Mobile application
- Real-time detection
- Cloud deployment

17. Conclusion

This system provides a professional AI-based solution for brain tumor detection using CNN. It improves diagnostic efficiency, accuracy, and supports healthcare professionals in decision-making.

18. REFERENCES

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.